

Melvin Thomas

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PROFILE

- Synthetic organic chemist with 4 years of experience in research and project management
- Skilled in organic synthesis, chromatographic purification, and characterization techniques
- Excellent communication skills gained from interdisciplinary team meetings and teaching experience over 2.5 years

TECHNICAL SKILLS

- **NMR:** ¹H NMR, ¹³C NMR, NMR yield calculation, 2D experiments
- **Organic Synthesis:** Glovebox, Schlenk line, liquid-liquid extraction, Column chromatography
- **Flow Reactor:** Setting up reactions, packing reactor with heterogeneous catalyst, preparing substrate solutions and cleaning reactor
- **Instrumentation Operation and Interpretation:** UV, FT-IR, GC-MS
- **Software:** ChemDraw, SciFinder, Reaxys, Mnova, Chimera, Endnote, MS Office

RESEARCH AND PROJECT MANAGEMENT EXPERIENCE

Graduate Student Researcher, Chemistry Department, UCSB Sept. 2019-Present

Thesis Project: Azacycle synthesis and ester amidation in flow

- Collaborated with industry partners on flow chemistry projects resulting in 2 publications (in progress)
- Presented progress in biweekly meetings and innovated ideas future directions of projects
- Synthesized Moclobemide (antidepressant) in flow reactor using a novel heterogeneous catalyst

Project: Developed synthetic methodologies to be environmentally responsible

- Discovered green protocol for deoxygenation of ketones in aqueous micelles
- Optimized palladium on carbon hydrogenation of nitroarenes in water

Master's Researcher, Mahatma Gandhi University, India Aug. 2016-Sept. 2017

- Developed methods to synthesize analogs of bioactive molecules (e.g, Crispine A, Harmicine)
- Isolated and characterized Garcinia acid from the fruit of *Garcinia Cambogia*

Research Intern, Juniata College, Pennsylvania June 2016-Aug. 2016

- Synthesized chiral aziridines by copper hydride mediated asymmetric reduction of aryl 2H- azirines
- Contributed to solving problems and presented findings at a research conference

SCIENCE COMMUNICATION EXPERIENCE

Graduate Teaching Assistant, UCSB Sept. 2019-Present

- Instructed organic chemistry lab courses by explaining reaction mechanisms and reactivity concepts
- Trained students to perform experiments and maintain safety practices

Film Director, Science Documentary Dec. 2016-Feb. 2017

- Led efforts to create a film on climate change for public awareness which received a Bronze Beaver Award at the National Science Film Festival

EDUCATION

M.S. Chemistry, University of California, Santa Barbara (UCSB), CA, USA Sept. 2022

B.S./M.S. Chemical Sciences, Mahatma Gandhi University, India July 2017

- **Awards:** S.N. Bose Internship Fellowship, Indian Academy of Science Research Fellowship

SELECTED PUBLICATIONS

- "Direct catalytic synthesis of N-substituted azacycles from cyclic ethers" (*manuscript under preparation*)
- "Heterogeneous catalytic amidation of esters in flow" (*manuscript under preparation*)

